Representation of Characters and Numbers, Basic Boolean Logic

OBJECTIVES

- The purpose of this applied session is to work on number systems conversion, apply Boolean logic, logic gates to build a circuit in Logisim

INSTRUCTIONS

- Download Logisim evolution from Moodle
- You may work in a small group.

Activity 1: Representation of Characters (using ASCII) and Numbers

In our computers, characters are represented using ASCII/Unicode and numbers are represented using 2's complement representation.

Task 1.1: Representation of Characters.

If your computer uses 7-bit ASCII, how would the computer represent the string "FA5"?

Task 1.2: Integer Numbers (4-bit) Representing Unsigned, Signed, 1's & 2's complement representations.

Create a table with all possible 4-bit binary values in one column and the (equivalent) decimal they represent in different columns using the following different notations:

(i) unsigned integer,
(ii) sign-magnitude notation,
(iii) 1's complement representation,
(iv) 2's complement representation.

4-bit binary	unsigned binary number	sign-magnitude representation	1's complement representation	2's complement representation
0000	0	+0	??	?
0001	1	+1	?	?

……	…..	……	……	……
1111	15	-7		

Task 1.3: Integer Numbers (8-bit)

(a) Assume you have 8 bits to represent binary numbers. What decimal value does the binary number 10110101_2 have in the different notations?
   (i) unsigned binary number
   (ii) sign-magnitude notation
   (iii) 1's complement
   (iv) 2's complement

(b) Represent the number -92_{10} in
   (i) 8-bit signed magnitude
   (ii) 8-bit 2's complement

Task 1.4: Addition and subtraction of signed numbers.

(a) Perform the following tasks using binary arithmetic with 8-bit 2's complement representation:
   (i) 33 + 92
   (ii) 33 - 92

(b) Convert the following numbers to 6-bit 2's complement notation and add them:
   (i) -16 and 11
   (ii) 16 and -11

Task 1.5: Concept of "Ranges of Numbers and Overflow".

(a) What are the ranges of numbers that can be represented in a computer if it is using:
   (i) 4-bit 2's complement representation
   (ii) 6-bit 2's complement representation
   (iii) 8-bit 2's complement representation

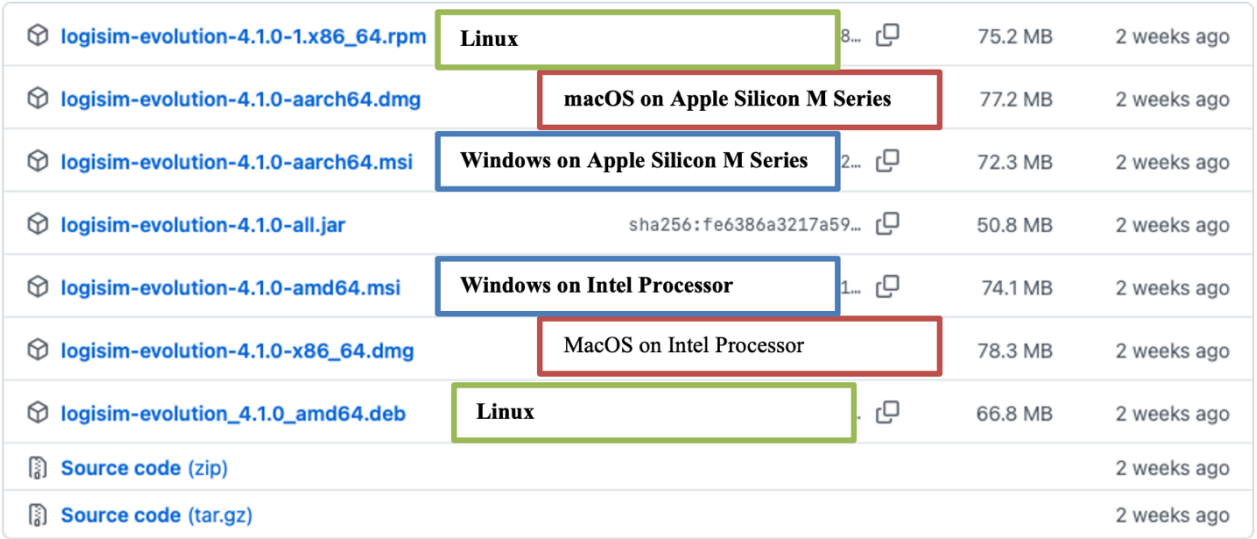
(b) What happens if you try to calculate 92+92 using binary arithmetic with 8-bit 2's complement representation? Can the result of this operation be accepted? How would you explain this operation referring to the "ranges of numbers" in 8-bit 2's complement representation?

Activity 2: Boolean Logic, Logic Gates and Logisim

Install Logisim software (link is also available in Moodle).

https://github.com/logisim-evolution/logisim-evolution/releases

Install the correct distribution according to your system architecture and OS.



Only for macOS: You may need to “Allow” installation under “Security & Privacy”.

Task 2.1: Using Logisim

Make yourself familiar with Logisim by doing a few very easy tasks:

- place one AND gate, 2 inputs and one output
- connect them
- poke the inputs to try simulation
- do the same with OR and NOT
- try some combinations of gates

Task 2.2: Build logic circuits using Logisim

In addition to AND, OR and NOT, there are a number of other Boolean logic operators. Examples are XOR (the exclusive OR), and NAND (a negated AND) operation. Their truth tables are shown in the tables below.

A	B	$A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

(a) XOR

A	B	$\overline{AB}$
0	0	1
0	1	1
1	0	1
1	1	0

(b) NAND

Build XOR and NAND logic in Logisim only using AND, OR, and NOT gates. Use the simulation to test your result.

Post-class Activity: Reflection

Complete your weekly reflection activity in the Ed Lesson. You would be able to copy/paste your reflections (**at least 200 words in each week**) to meet assignment hurdle requirement.